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KALPATARU INSTITUTE OF TECHNOLOGY TIPTUR

“ENHANCING ACCIDENT PREDICTION THROUGH INTEGRATED KNN AND DBSCAN ALGORITHMS FOR SUPERIOR ACCURACY”

INFORMATION SCIENCE AND ENGINEERING

MAJOR PROJECT PHASE-2(BIS786)

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PROJECT TITLE

"ENHANCING ACCIDENT PREDICTION THROUGH INTEGRATED KNN AND DBSCAN ALGORITHMS FOR SUPERIOR ACCURACY"

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ABSTRACT

- Road traffic accidents are a growing global concern, endangering lives and infrastructure. Accurate prediction of accident severity is essential for effective emergency response and safety planning. This study introduces a hybrid model integrating KNN and DBSCAN algorithms improve prediction accuracy. The combined approach enhances reliability, reduces noise sensitivity, and supports safer urban mobility.

INTRODUCTION

- Road traffic accidents are a major global concern, causing significant loss of life and economic damage each year. Accurate prediction of accident severity can help improve emergency response and safety planning. Traditional models often fail to handle noisy and complex traffic data effectively. Machine learning provides a promising solution, but no single algorithm performs well across all data types. Therefore, this study integrates KNN and DBSCAN algorithms to enhance prediction accuracy and model reliability.

📋 OBJECTIVES OF THE STUDY

- ❑To develop an intelligent accident severity prediction system using Machine Learning.
- ❑To apply and analyse the performance of K-Nearest Neighbors (KNN) for severity classification.
- ❑To utilize DBSCAN for clustering accident data and detecting anomalies/noise.
- ❑To design a hybrid model (KNN + DBSCAN) that improves accuracy .
- ❑To contribute to safer urban mobility through data-driven accident prevention and management strategies.

PROBLEM STATEMENT

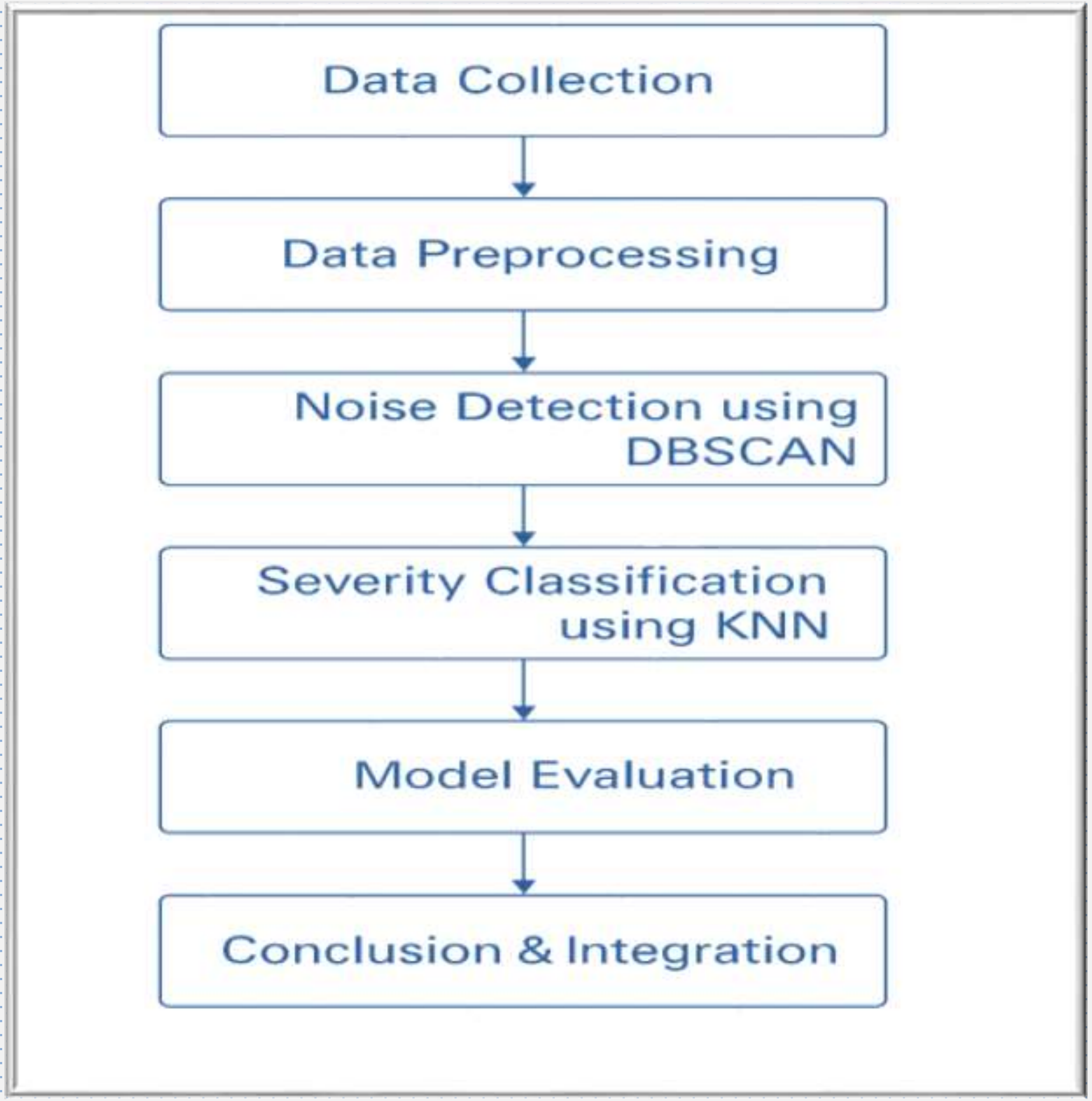
- Road traffic accidents cause significant loss of life and property, and predicting their severity remains a major challenge. Traditional models often fail to perform well due to noisy, imbalanced, and complex real-world datasets. Single algorithms like KNN or DBSCAN have limitations such as overfitting, poor handling of noise, or reduced accuracy. There is a need for a more reliable and accurate model that can effectively analyze accident data. Hence, this project proposes an integrated KNN–DBSCAN model to enhance accident prediction accuracy and robustness.

LITERATURE SURVEY

Author / Year	Methodology / Technology Used	Key Findings & Results	Limitations / Gaps
AlHashmi, 2024	Hybrid KNN + DBSCAN	Improved prediction accuracy; detects accident hotspots; handles spatial outliers	Tested on limited datasets; real-time deployment not explored
Briki et al., 2024	Cluster-based severity resampling + Classification	Addressed class imbalance; improved model performance	Focused mainly on dataset preprocessing; not real-time
Chakravarthi, 2025	Hybrid KNN-DBSCAN	Enhanced detection of clusters and outliers; better severity prediction	Requires careful parameter tuning; computationally intensive
Khosravi et al., 2024	KNN	High accuracy in accident severity prediction; simple to implement	Sensitive to noisy data; limited in handling spatial clustering


ROAD ACCIDENTAL
STATISTICS
IN INDIA, 2018–2025
4,49,002
TOTAL ACCIDENTS
1,52,853
FATALITIES
4,32,427
INJURIES

"Proposed Methodology for Accident Severity Prediction"



🧾 Proposed Methodology

Step 1: **Data Collection**

- Real-world accident dataset with features like location, weather, time, and severity.

Step 2: **Data Preprocessing**

- Handle missing values, normalize numerical data.
- Encode categorical variables.
- Remove irrelevant or redundant attributes.

Step 3: **Noise Detection using DBSCAN**

- DBSCAN identifies **dense clusters** and filters **outliers/noise**.
- Improves data quality before classification.

Step 4: **Severity Classification using KNN**

- **KNN classifies severity based on nearest neighbors.**
- **Works on DBSCAN-cleaned data for better accuracy.**

Step 5: **Model Evaluation**

- **Use metrics like accuracy, precision, recall, and F1-score.**
- **Compare hybrid model with standalone KNN and DBSCAN.**

Step 6: **Conclusion & Integration**

- **Final model supports predictive insights for accident management systems.**

Existing System vs Proposed System

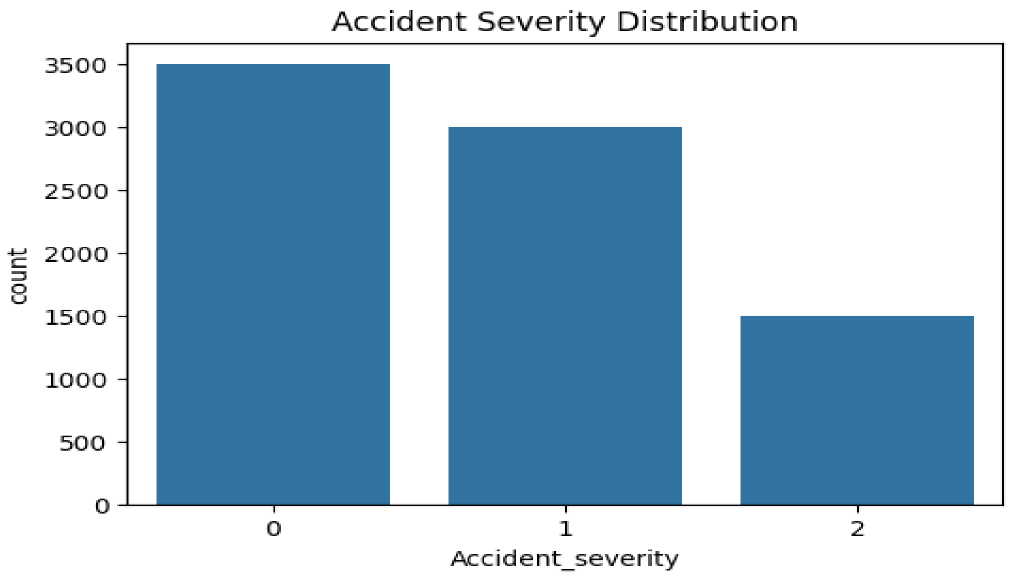
Criteria	Existing System	Proposed System
Approach	Uses individual machine learning algorithms such as KNN, Random Forest.	Integrates KNN and DBSCAN algorithms to form a hybrid model for better prediction accuracy and noise handling.
Data Handling	Sensitive to noise, outliers, and imbalanced datasets, leading to reduced model reliability.	DBSCAN removes noise and detects dense clusters, improving data quality before classification.
Accuracy	Moderate accuracy due to limitations of single models and lack of data preprocessing.	Achieves higher accuracy
Model Type	Single algorithm-based (supervised or unsupervised).	Hybrid model combining supervised (KNN) and unsupervised (DBSCAN) learning.
Scalability	Limited scalability and poor performance on large, heterogeneous datasets.	Improved scalability with
Real-Time Application	Struggles to handle real-time or dynamic data updates.	Can be adapted for real-time prediction and alert systems using pre-clustered data.
Result	Inconsistent and less reliable predictions	More consistent, accurate, and interpretable results that support safer urban mobility.

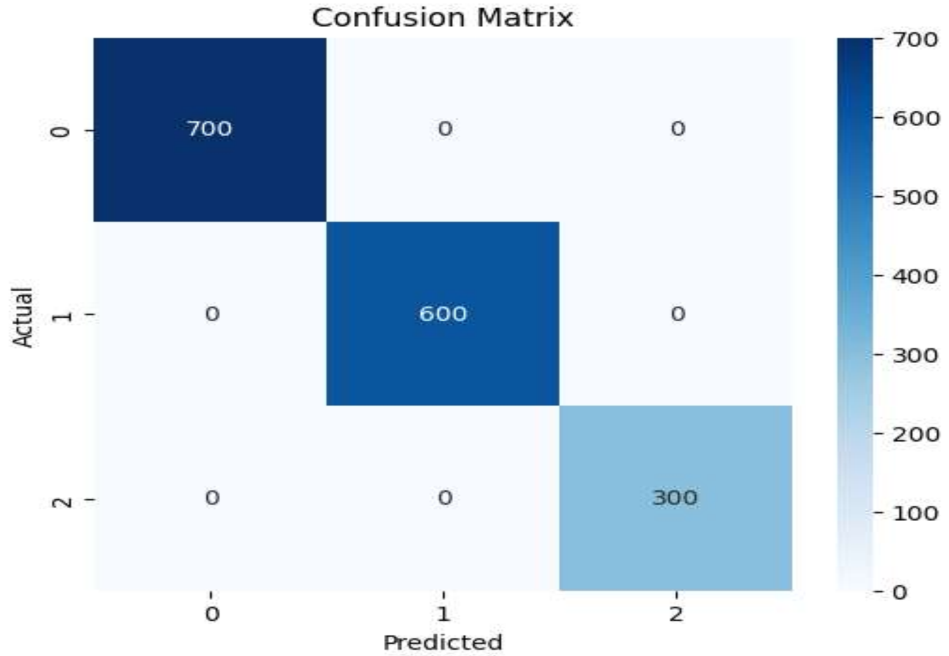
Advantages:	Disadvantages:
• Predicts accidents more accurately than using KNN or DBSCAN alone. • Can handle noisy or unusual data well. • Works on different types of traffic data. • Helps with faster emergency response and resource planning. • Can use sensor data for smarter traffic safety.	• More complex to set up and run. • Needs more computer power for large data. • DBSCAN parameters need careful tuning. • Harder to use in real-time traffic systems. • May need changes for different cities or road conditions.

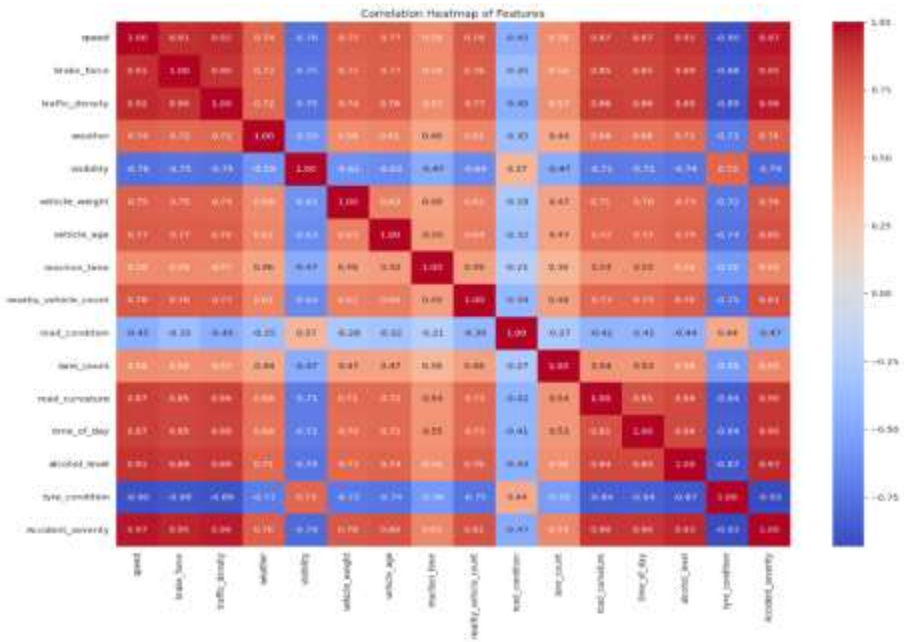
OUTCOMES

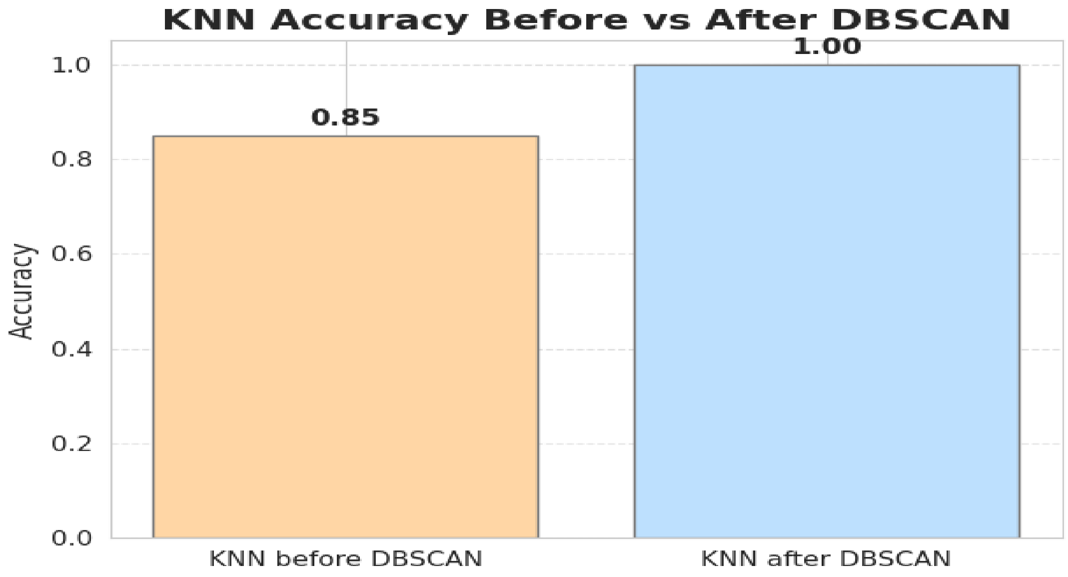
- Increased prediction accuracy
- Better noise handling via DBSCAN
- Enhanced classification through KNN
- Support for traffic authorities with data insights
- Improved real-time accident detection
- Support for traffic authorities with data insights
- Potential for integration into smart traffic systems

RESULT


Accident Severity Distribution
count
Accident_severity
0
1
2
3500
3000
2500
2000
1500
1000
500

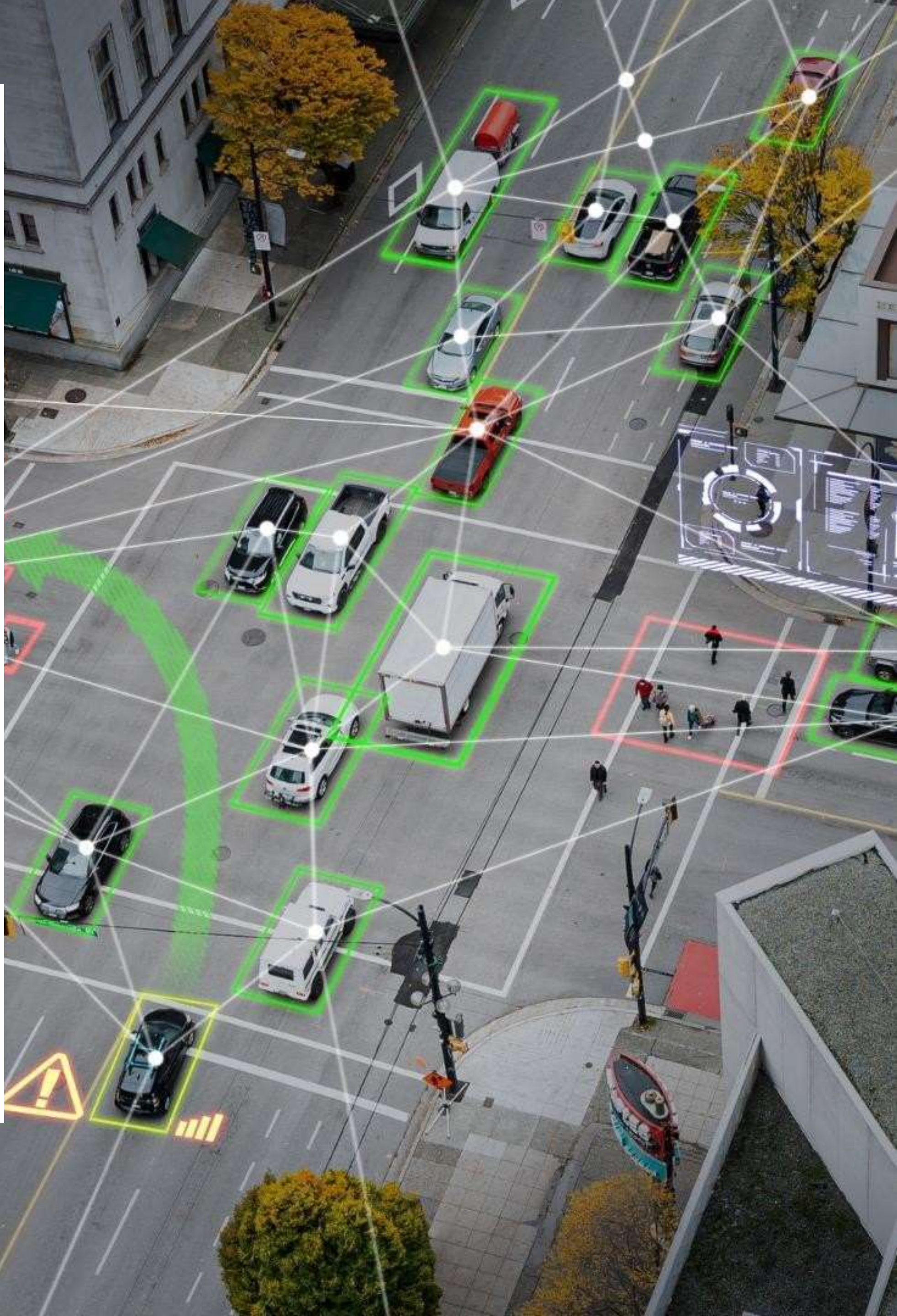

Confusion Matrix
Actual
Predicted
700
600
300
0


Correlation Heatmap of Features


KNN Accuracy Before vs After DBSCAN
Accuracy
0.85
1.00
KNN before DBSCAN
KNN after DBSCAN

Future Work

- Exploring other algorithms for traffic prediction
- Enhancing the preprocessing techniques
- Investigating hybrid approaches combining KNN, DBSCAN, etc
- Extending the framework to other cities or regions
- Integrating additional data sources (e.g., weather information)



CONCLUSION

- The hybrid KNN-DBSCAN model can predict how serious accidents might be by combining classification and clustering methods. Tests show it works better than using KNN or DBSCAN alone, handling noisy data well and giving accurate results on different traffic datasets. This method can help improve emergency response and manage resources more efficiently. Adding data from IoT sensors can make traffic safety even better. In the future, the model will be tested for real-time use, on larger datasets, and in different locations.

- In summary, this hybrid model doesn’t stop accidents directly, but it helps us **understand when and where accidents are more likely** to occur based on past patterns. This can be used to issue **travel safety warnings** and **improve road planning and policies**.

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Thank you